

(Table 1). These tools help identify vulnerabilities and potential security issues in your smart contracts.

Installation and working with Slither for security audit and vulnerability testing

Slither is a powerful and versatile tool that aids in identifying vulnerabilities and improving the security of Ethereum smart contracts. Its features make it an essential component of the development toolkit for those working in the blockchain space.

Slither is an open source tool designed to analyse Ethereum smart contracts for security vulnerabilities and issues. It's widely used by developers and auditors to ensure the safety and reliability of smart contracts.

Key features of Slither

Slither is like a shield against code vulnerabilities. Let's delve deeper into some of its key features and witness how it helps secure smart contract development.

Static analysis: Slither performs static analysis on smart contracts, which means it analyses the code without executing it. This helps identify potential vulnerabilities and issues without the need to deploy the contract on the blockchain.

Vulnerability detection: One of Slither's primary functions is to detect a wide range of vulnerabilities and weaknesses in smart contracts. These vulnerabilities can include issues like reentrancy attacks, unchecked send calls, integer overflows, and more.

Custom rules: Slither allows users to define custom rules and checks that are specific to their project's requirements. This flexibility enables developers to target specific patterns or issues that might not be covered by default rules.

Integration with build process: It can be integrated into the development workflow, often as part of the continuous integration (CI) pipeline, to automatically analyse

contracts whenever changes are made to the codebase. This ensures that potential issues are caught early in the development process.

Interactive reports: Slither generates comprehensive reports that detail the identified vulnerabilities and issues. The reports provide insights into the location of the issue, its severity, and possible solutions. This information assists developers in understanding and fixing the problems.

Gas estimations: For Ethereum smart contracts, gas costs are a critical consideration. Slither provides gas estimations for various execution paths within the contract, helping developers understand the potential gas expenses of their code.

Support for multiple languages: While Slither is mainly designed for Solidity, the primary programming language for Ethereum smart contracts, it also supports other languages that compile to Ethereum bytecode, such as Vyper.

Continuous updates: The tool is actively maintained by the community, which means it receives regular updates to improve its detection capabilities and support the evolving landscape of smart contract security issues.

To install Slither, use pip installer:

```
pip install slither-analyzer
```

After completing the installation, you can verify it as follows:

```
slither --version
```

Once Slither is successfully installed, it can be used to analyse Solidity smart contracts for security vulnerabilities.

For example, to analyse a Solidity file named 'MyContract.sol', you can navigate to its directory and then run:

```
slither MyContract.sol
```

Slither will analyse the contract

and provide you with a report of any potential security issues it detects.

Let's say there is a Solidity smart contract named 'SimpleToken.sol':

```
pragma solidity ^0.8.0;
contract SimpleToken {
    mapping(address => uint256) public
    balances;

    constructor() {
        balances[msg.sender] = 1000;
    }
    function transfer(address to, uint256
    amount) public {
        require(balances[msg.sender] >=
        amount, "Insufficient balance");
        balances[msg.sender] -= amount;
        balances[to] += amount;
    }
}
```

To perform an audit using Slither, follow these steps:

- Open a terminal window.
- Navigate to the directory containing your Solidity contract.
- Run Slither on your contract

```
> slither SimpleToken.sol
```

Slither will analyse your Solidity code and provide a report on the potential security issues it detects.

Slither static analysis report

Summary :

```
High : 0
Medium : 0
Low : 0
Informational : 1
```

Detectors :

```
UninitializedState : uninitialized
state variables
  · contracts/SimpleToken.sol : L3 : 3
| SimpleToken | balances
```

In this example, Slither's uninitialised state detector detected that the balances mapping is uninitialised